INORGANIC CHEMISTRY CUMULATIVE EXAM

February 7, 1998

A variety of interesting features of transition metal chemistry are exemplified by the behavior of copper and its derivatives. Some of these examples are indicated below. Answer any 6.

1. Aqueous solutions of copper(II) are blue. What electronic transition(s) give rise to the color?
2. Addition of ammonia to these solutions result in the formation of deep blue color? What is going on chemically and electronically?
3. The kinetic studies of copper(II) substitution indicate that ligand exchange occurs with rates larger than most of the other transition metals. Why?
4. Complexes such as bis(2,4-pentandionato)copper(II) are almost white when dry and turn green or blue when exposed to moist air. Explain.
5. When copper sulfate is incorporated into a KBr pellet, the pellets slowly turn black. Explain.
6. In the manufacture of printed circuit boards, the copper foil is selectively removed by dissolving the unwanted copper with concentrated solutions of ferric chloride. What is the chemistry involved in this reaction? Why not use hydrochloric acid or sulfuric acid?
7. In the past decade, a major discovery was made which involved copper compounds. The potential benefits of this discovery are far reaching but practical considerations have limited the application of these materials. What was the discovery? What form of copper was involved?